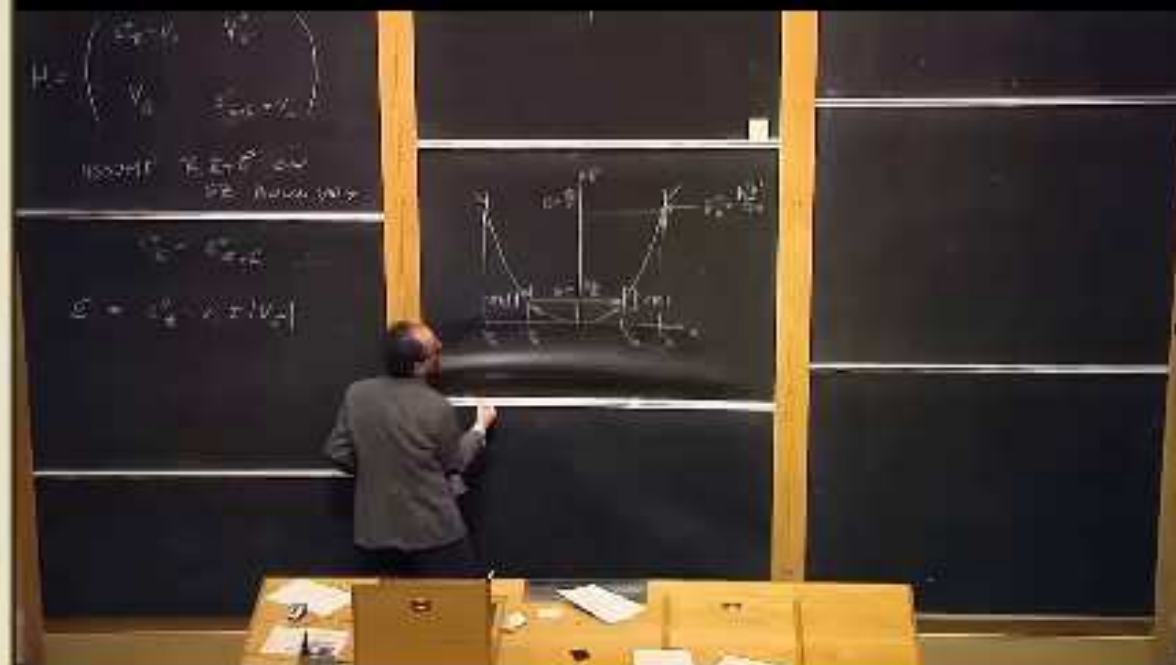




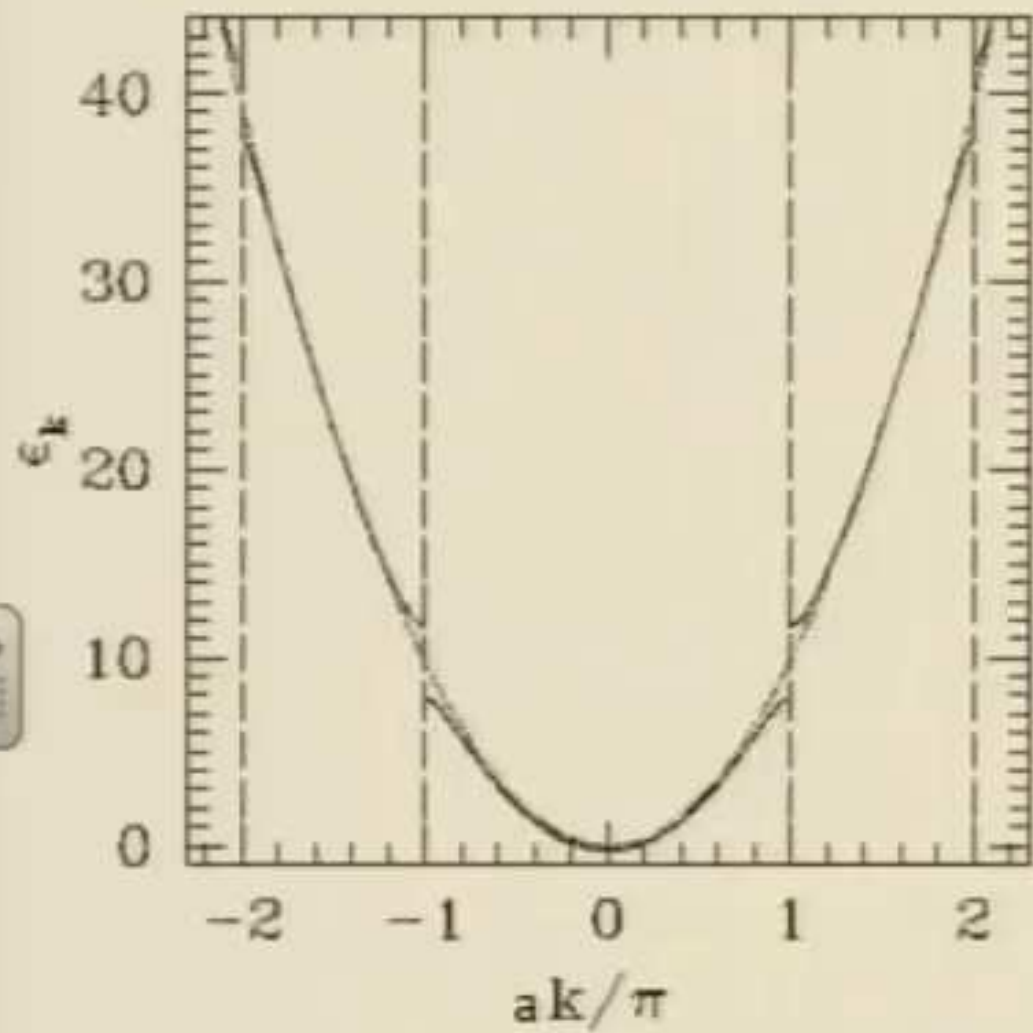
Slides

Condensed Matter Physics

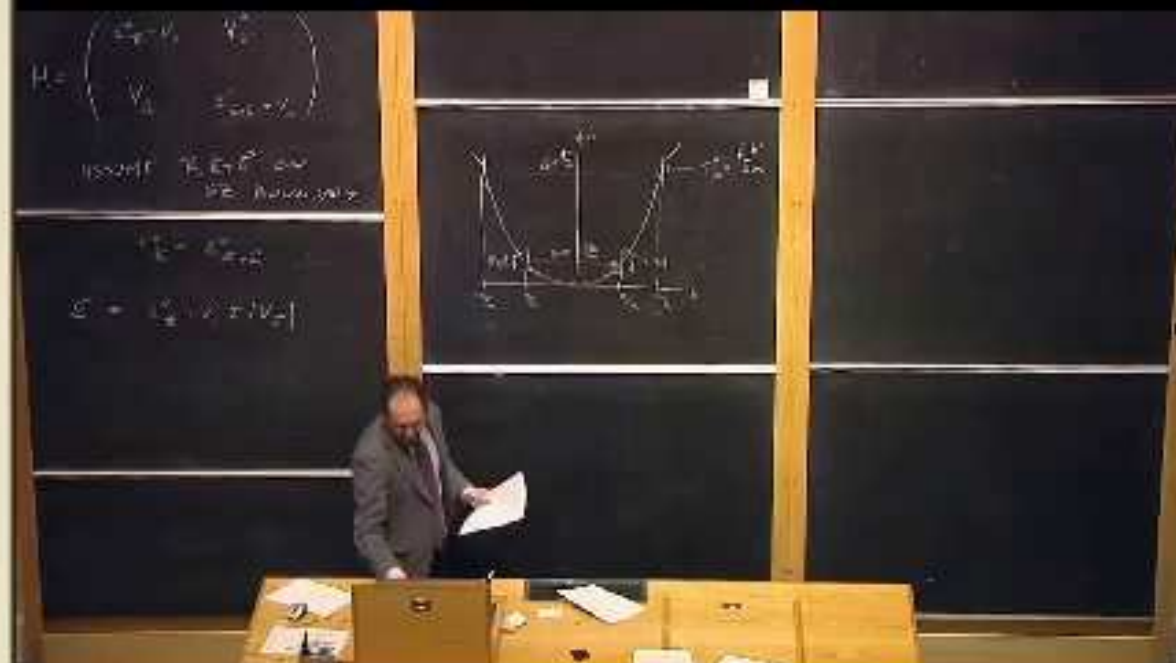
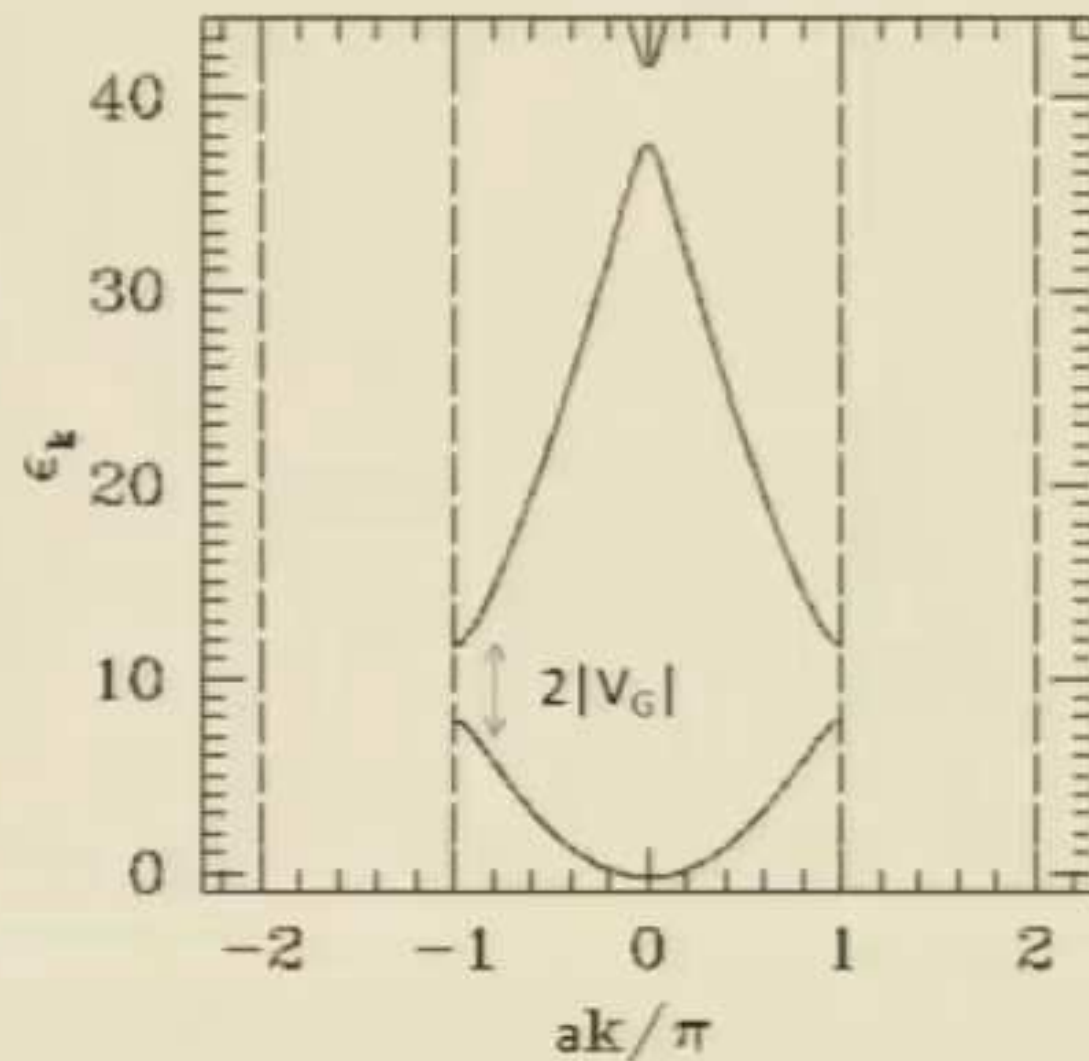
Lecture 15



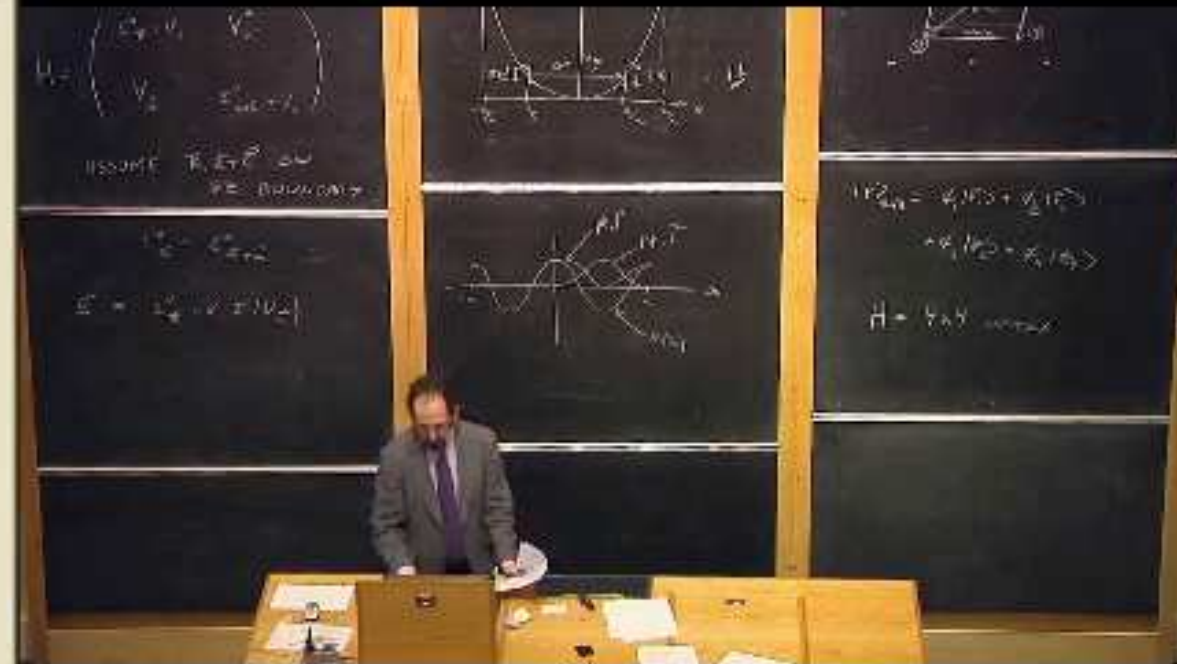
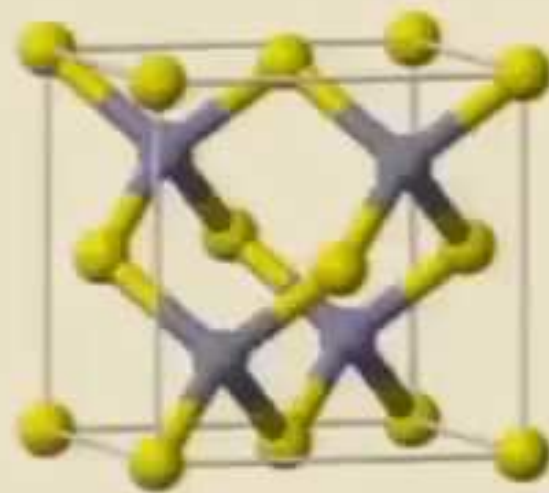
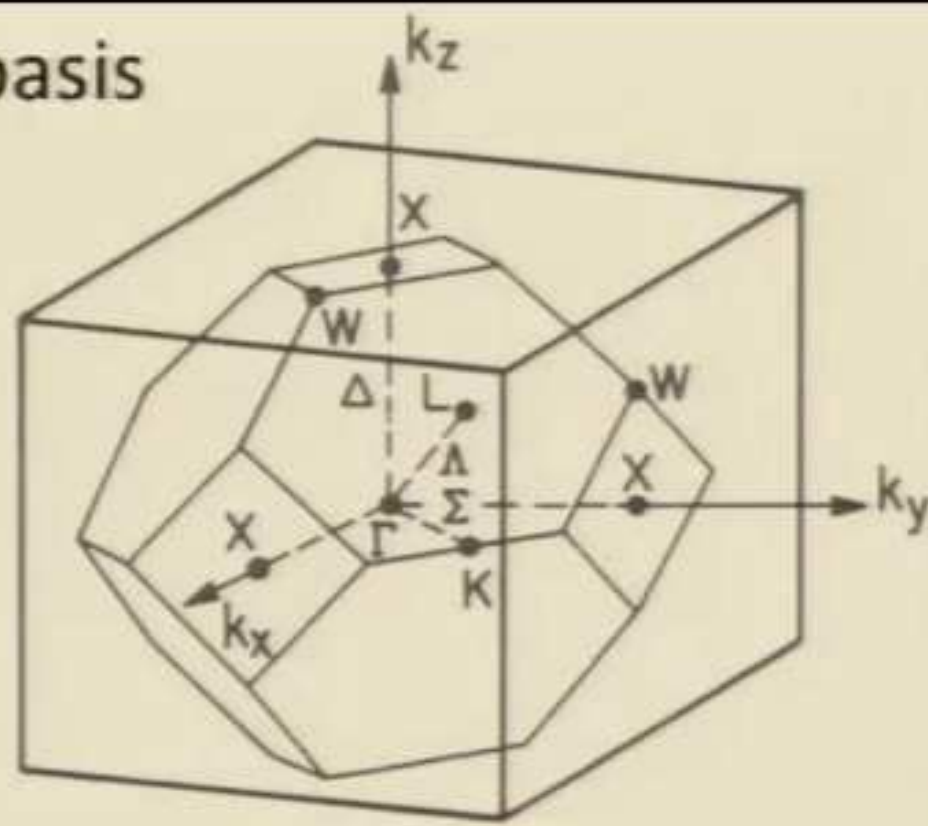
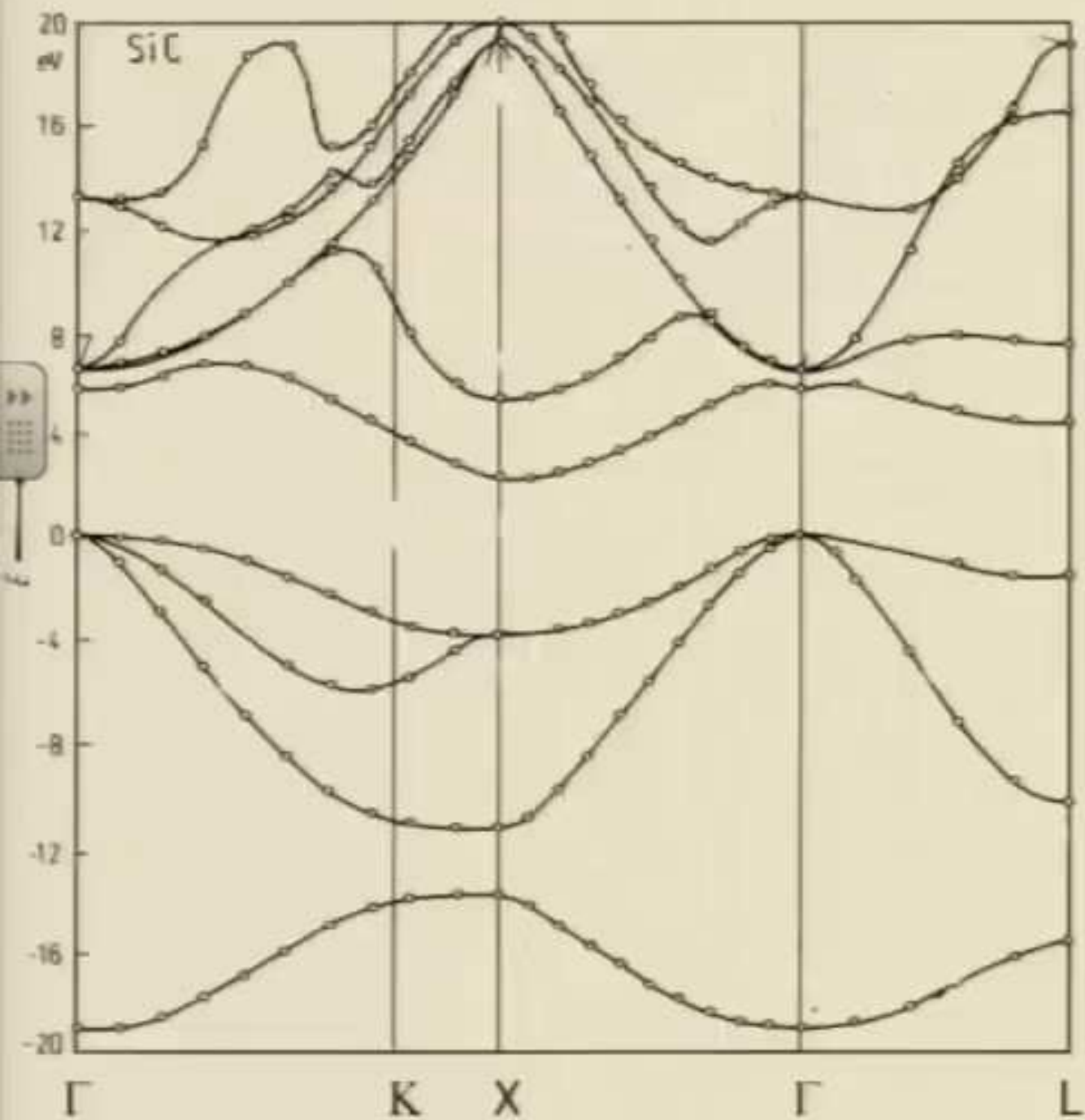
d=1 extended zone



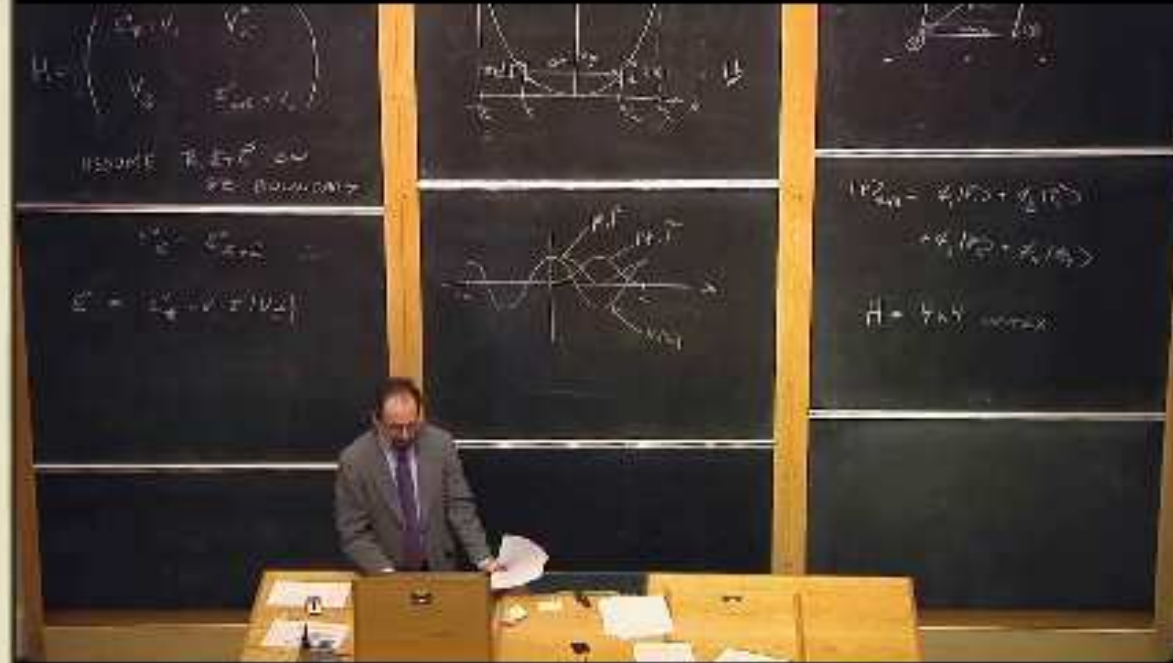
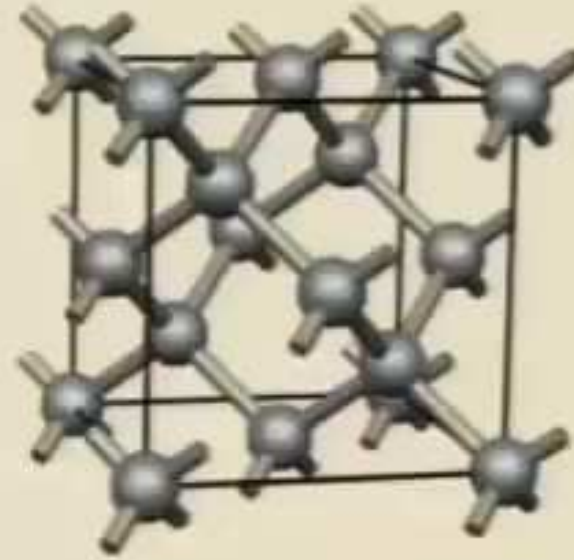
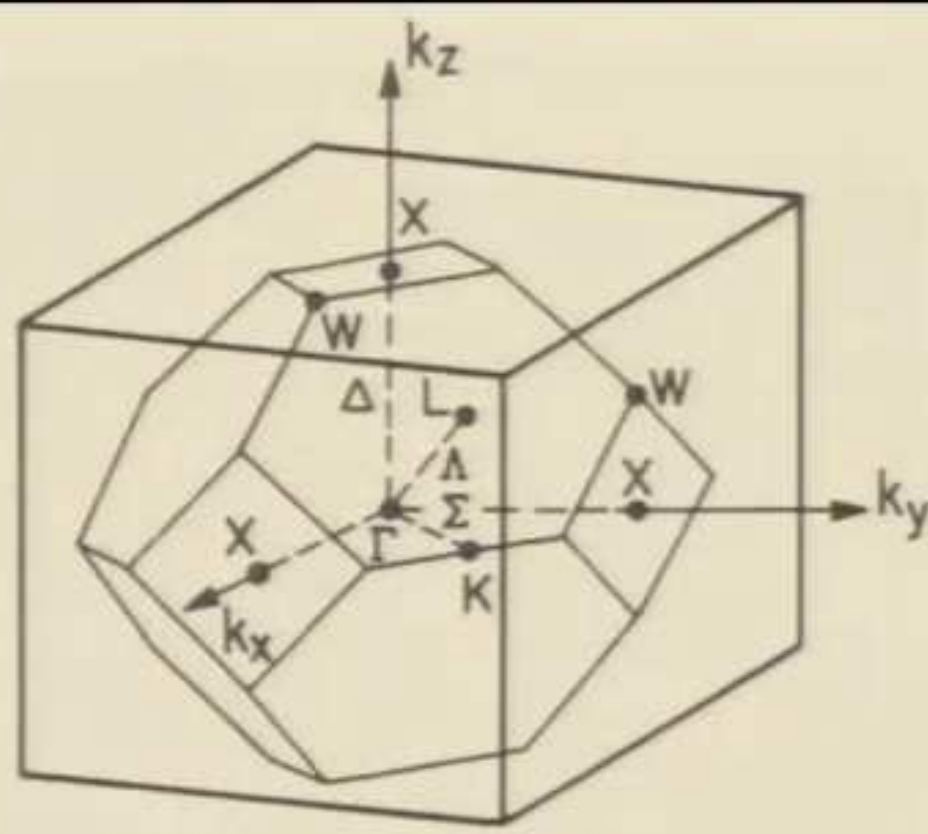
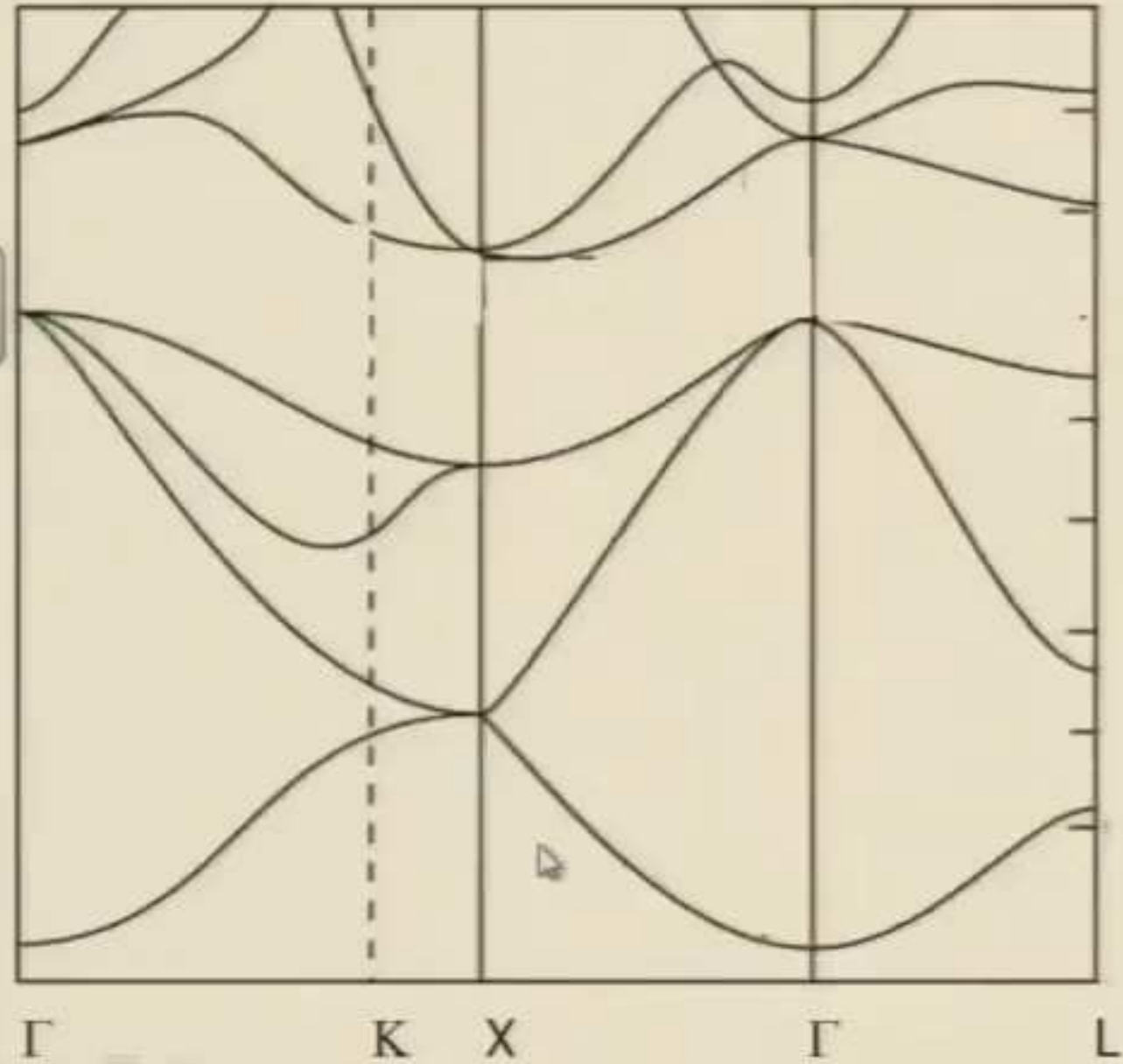
d=1 reduced zone



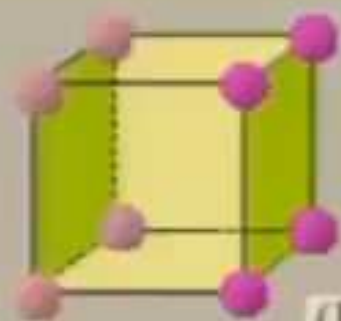
Silicon Carbide = FCC with a 2-atom basis
 Si @ $[0,0,0]$ and C @ $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$



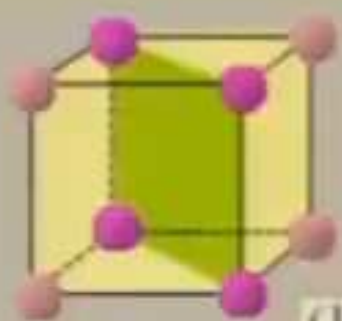
Silicon = FCC with a 2-atom basis
 Si @ $[0,0,0]$ and Si @ $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$



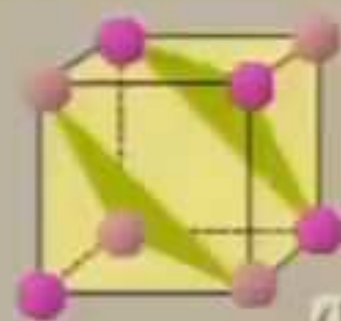
SC



(100)

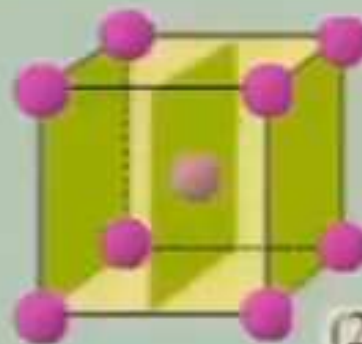


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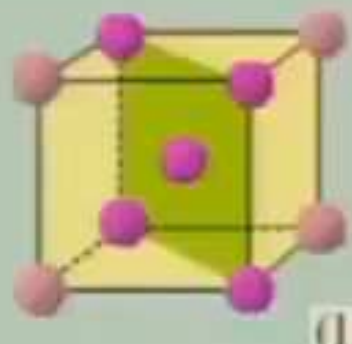


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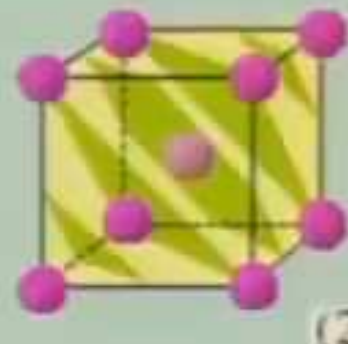
bcc



(200)

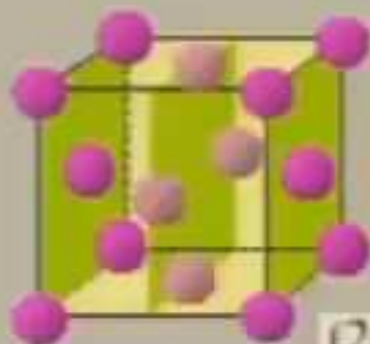


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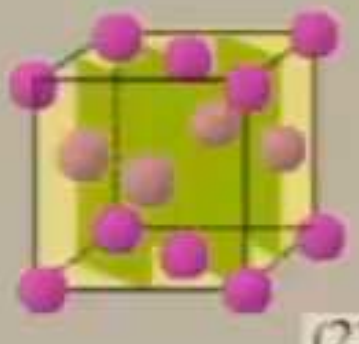


(222)

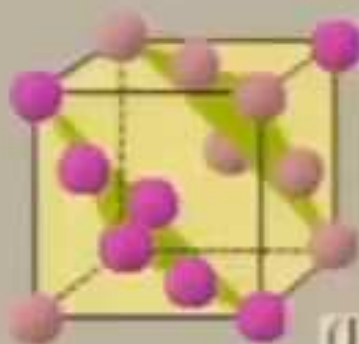
fcc



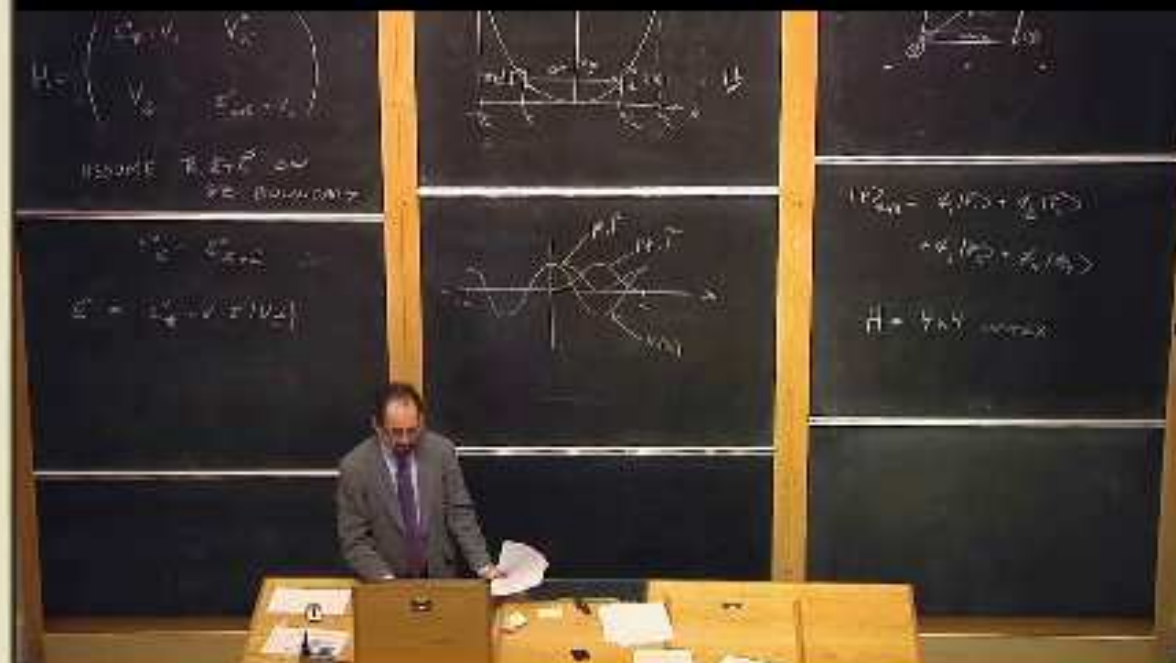
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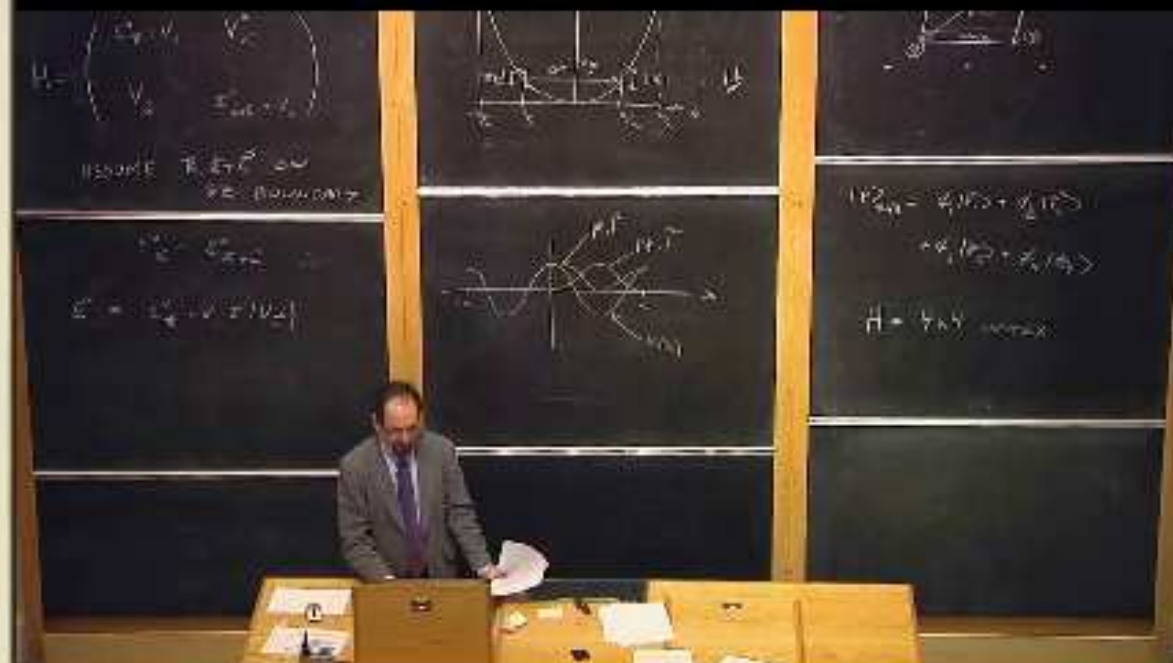
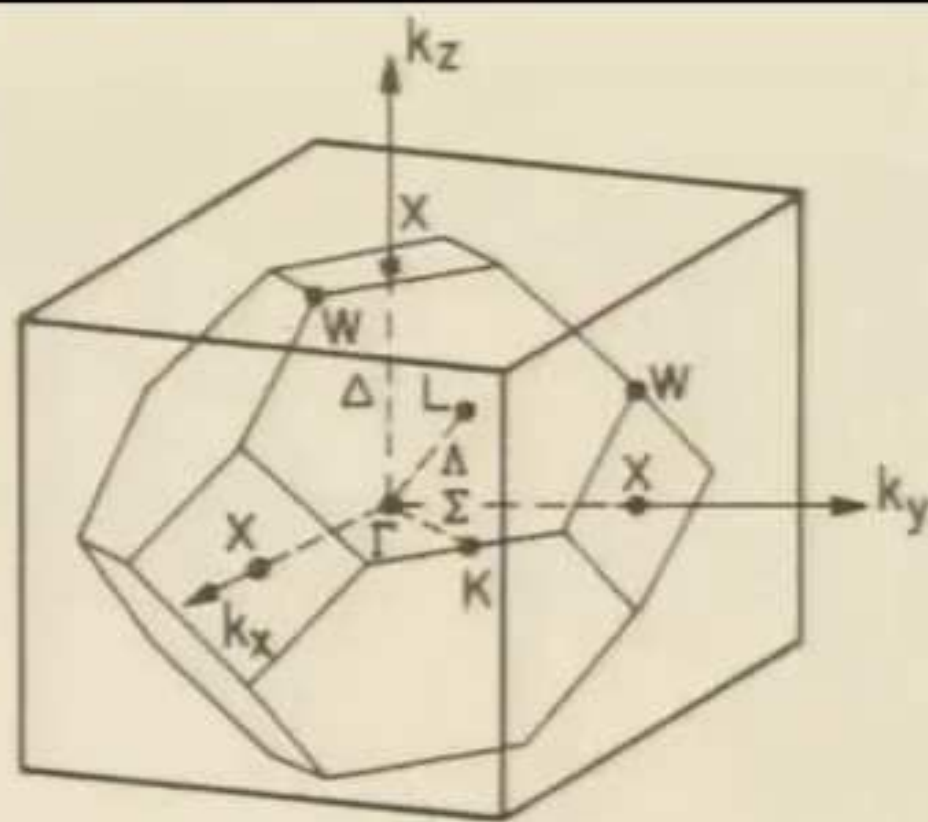
(220)



(111)



1st Brillouin Zone of an FCC lattice
 =same shape as Wigner Seitz
 cell of a BCC lattice



Silicon = FCC with a 2-atom basis
 Si @ $[0,0,0]$ and Si @ $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$

